THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 8 · SET 02

Max. Marks: 60 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Multiple Choice Questions (12 × 1 = 12) — **1 questions · 12 marks**

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) The number of non-square natural numbers between 9^2 and 10^2 is:

(a) 17 (b) 19 (c) 20 (d) 18

Ans: (d) Between n^2 and $(n+1)^2$ there are $2n$ non-square numbers: $2 \times 9 = 18$.

(ii) Which of the following is a perfect cube?

(a) 1800 (b) 2025 (c) 1728 (d) 3000

Ans: (c) $1728 = 12^3$ ($2^6 \times 3^3$).

(iii) The value of 2^{-4} is:

(a) 1/16 (b) −16 (c) −8 (d) 1/8

Ans: (a) $2^{-4} = 1/2^4 = 1/16$.

(iv) $(3^5)^2 \div 3^7$ equals:

(a) 3 (b) 27 (c) 9 (d) 81

Ans: (b) $3^{10} \div 3^7 = 3^3 = 27$.

(v) The standard form of 6,50,000 is:

(a) 65×10^4 (b) 6.5×10^4 (c) 6.5×10^5 (d) 0.65×10^5

Ans: (c) $650000 = 6.5 \times 100000 = 6.5 \times 10^5$.

(vi) 449 written in Roman numerals is:

(a) CDXLIX (b) CCCXLIX (c) CDLIX (d) CDXLVIIII

Ans: (a) 400 = CD, 40 = XL, 9 = IX ⇒ CDXLIX.

(vii) The base-3 number 212 equals which decimal number?

(a) 21 (b) 22 (c) 23 (d) 25

Ans: (c) $2\times9 + 1\times3 + 2\times1 = 23$.

(viii) A quadrilateral with exactly one pair of parallel sides is a:

(a) Parallelogram (b) Trapezium (c) Rhombus (d) Rectangle

Ans: (b) By definition, a trapezium has exactly one pair of parallel sides.

(ix) In an isosceles trapezium ABCD with AB ∥ DC, if ∠A = 70°, then ∠D is:

(a) 70° (b) 110° (c) 90° (d) 100°

Ans: (b) ∠A + ∠D = 180° (co-interior angles) ⇒ ∠D = 110°.

(x) Kaprekar's constant for 4-digit numbers is:

(a) 6174 (b) 495 (c) 1089 (d) 9999

Ans: (a) The routine on 4-digit numbers (not all digits same) always ends at 6174.

(xi) $(a + b)^2 - (a - b)^2$ equals:

(a) 2ab (b) $2a^2 + 2b^2$ (c) 0 (d) 4ab

Ans: (d) $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$.

(xii) If 4 pencils cost ₹ 30, the cost of 10 pencils is:

(a) ₹ 70 (b) ₹ 80 (c) ₹ 75 (d) ₹ 120

Ans: (c) 4 : 30 = 10 : x ⇒ x = 300/4 = 75.

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the smallest number by which 72 must be multiplied to get a perfect cube.

$72 = 2^3 \times 3^2$ (1 mark)
3^2 needs one more 3 to make 3^3 ⇒ multiply by 3 (product 216 = 6^3) (1 mark)

3. Evaluate: $(2^{-1} + 3^{-1})^{-1}$

$2^{-1} + 3^{-1} = 1/2 + 1/3 = 5/6$ (1 mark)
$(5/6)^{-1} = 6/5$ (1 mark)

4. Convert $(110101)_2$ into a decimal number.

1×32 + 1×16 + 0×8 + 1×4 + 0×2 + 1×1 (1 mark)
= 53 (1 mark)

5. Without actual division, check whether 234567 is divisible by 9.

Digit sum = 2+3+4+5+6+7 = 27 (1 mark)
27 is divisible by 9 ⇒ 234567 is divisible by 9 (1 mark)

OR

Find the digit * so that 3*76 is divisible by 11.

(Sum of digits at odd places) − (sum at even places) = (3+7) − (*+6) = 4 − * (1 mark)
This must be 0 ⇒ * = 4 (3476 = 11 × 316) (1 mark)

6. The diagonals of a rectangle are (3x + 5) cm and (5x − 7) cm. Find x and the length of each diagonal.

Diagonals of a rectangle are equal: 3x + 5 = 5x − 7 ⇒ x = 6 (1 mark)
Each diagonal = 3(6) + 5 = 23 cm (1 mark)

7. Divide ₹ 540 between Meena and Neha in the ratio 4 : 5.

Total parts = 9; 1 part = ₹ 60 (1 mark)
Meena = ₹ 240, Neha = ₹ 300 (1 mark)

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Convert 69 into (i) base 5 and (ii) base 3.

(i) 69 = 2×25 + 3×5 + 4 ⇒ $(234)_5$ (1½ marks)
(ii) 69 = 2×27 + 1×9 + 2×3 + 0 ⇒ $(2120)_3$ (1½ marks)

9. In a rhombus ABCD, ∠A = 70°. Find ∠B, ∠C and ∠D. State the property used.

Adjacent angles are supplementary ⇒ ∠B = 110° (1 mark)
Opposite angles are equal ⇒ ∠C = 70°, ∠D = 110° (1 mark)
Property: opposite angles of a parallelogram (rhombus) are equal and adjacent angles are supplementary (1 mark)

OR

In trapezium PQRS, PQ ∥ SR, ∠P = 75° and ∠R = 110°. Find ∠S and ∠Q.

> Co-interior angles between parallel sides PQ and SR: ∠P + ∠S = 180° ⇒ ∠S = 105° (1½ marks)
> ∠Q + ∠R = 180° ⇒ ∠Q = 70° (1½ marks)

10. Expand and simplify:
(i) $(2a + 3)(a^2 - 4a + 5)$ (ii) $(3x - 2y)(x + 5y)$

> (i) $2a^3 - 8a^2 + 10a + 3a^2 - 12a + 15 = 2a^3 - 5a^2 - 2a + 15$ (2 marks)
> (ii) $3x^2 + 15xy - 2xy - 10y^2 = 3x^2 + 13xy - 10y^2$ (1 mark)

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 1176 must be multiplied so that the product is a perfect square. Find the square root of the product.
(ii) The area of a square field is 2704 m². Find the length of its side.

> (i) $1176 = 2^3 \times 3 \times 7^2$ (1 mark); unpaired factors 2 and 3 ⇒ multiply by $2 \times 3 = 6$ (1 mark)
> Product $7056 = 2^4 \times 3^2 \times 7^2$, $\sqrt{7056} = 4 \times 3 \times 7 = 84$ (1 mark)
> (ii) $2704 = 2^4 \times 13^2$ ⇒ side $= 4 \times 13 = 52$ m (2 marks)

OR

(i) Find the smallest number by which 2000 must be divided so that the quotient is a perfect cube. Find the cube root of the quotient.
(ii) The volume of a cube is 4096 cm³. Find the length of its edge.

> (i) $2000 = 2^4 \times 5^3$ (1 mark); divide by 2 (1 mark); quotient 1000, cube root 10 (1 mark)
> (ii) $4096 = 2^{12}$ ⇒ edge $= 2^4 = 16$ cm (2 marks)

12. If $17^2 = 289$, find: (i) 1.7^2 (ii) 0.17^2 (iii) 170^2 (iv) 1700^2 (v) 0.017^2

> (i) 2.89 (ii) 0.0289 (iii) 28900 (iv) 2890000 (v) 0.000289 (1 mark each)

OR

Simplify and write in exponential form: (i) $5^{-2} \times 5^7$ (ii) $2^{-3} \div 2^{-5}$ (iii) $(9^{-3})^2$ (iv) $(3/4)^{-2}$ (v) $p^6q^2 \times (pq)^{-2}$

> (i) 5^5 (ii) 2^2 (iii) 9^{-6} (iv) $(4/3)^2 = 16/9$ (v) p^4 (1 mark each)

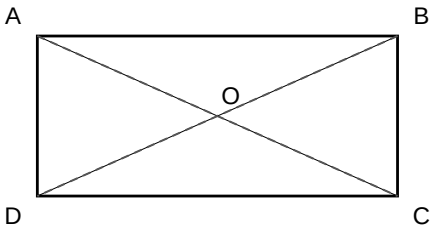
13. (i) Apply Kaprekar's routine to the 3-digit number 352 and show every step until it reaches 495.
(ii) What happens if the routine is applied to 111? Why?
(iii) Apply the routine to 2025 (4-digit) and state the number of steps taken to reach 6174.

> (i) 532 − 235 = 297; 972 − 279 = 693; 963 − 369 = 594; 954 − 459 = 495 (2 marks: 4 steps, each correct = ½ mark)
> (ii) 111 − 111 = 0: all digits are equal, so the routine collapses to 0 and never reaches 495 (1 mark)
> (iii) 5220 − 0225 = 4995; 9954 − 4599 = 5355; 5553 − 3555 = 1998; 9981 − 1899 = 8082; 8820 − 0288 = 8532; 8532 − 2358 = 6174 ⇒ 6 steps (2 marks)

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. A rectangular herb garden ABCD has a diagonal path AC = 26 cm on the school's model plan. The diagonals AC and BD meet at O, and the side AB = 10 cm.



(i) Find the length of OB.

Diagonals of a rectangle are equal and bisect each other: OB = ½ × 26 = 13 cm

(ii) Find the length of the side BC.

BC = $\sqrt{(26^2 - 10^2)} = \sqrt{576} = 24$ cm

(iii) Find the perimeter and the area of the rectangular garden.

Perimeter = 2(10 + 24) = 68 cm (1 mark); area = 10 × 24 = 240 cm² (1 mark)

OR

(iii) A string is laid along both diagonals at ₹ 45 per cm on the model. Find the total cost.

Length = 26 + 26 = 52 cm (1 mark); cost = 52 × 45 = ₹ 2340 (1 mark)

15. In a laboratory a culture starts with 5×10^4 cells, and the number of cells becomes 10 times larger every day.

(i) How many cells will there be after 3 days? Give your answer in standard form.

$5 \times 10^4 \times 10^3 = 5 \times 10^7$

(ii) Write the number of cells after 3 days in usual form.

5×10^7 = 5,00,00,000 (five crore)

(iii) Find the ratio of the number of cells after 2 days to that after 5 days.

After 2 days: 5×10^6; after 5 days: 5×10^9 (1 mark); ratio = $10^6 : 10^9 = 1 : 1000$ (1 mark)

OR

(iii) After how many days will the culture first exceed 5×10^9 cells?

Cells after n days = $5 \times 10^{4+n}$; equals 5×10^9 at n = 5 (1 mark), so it first exceeds this on day 6 (1 mark)

16. A school canteen prepares a sherbet by mixing lemon juice, sugar and water in the ratio 1 : 2 : 7 to make 5 litres of sherbet in each batch.

(i) How many litres of sugar are used in one batch?

Parts = 10; sugar = 2/10 × 5 = 1 litre

(ii) Write the ratio of lemon juice to water in simplest form.

1 : 7

(iii) How much water is needed along with 2.4 litres of lemon juice (same ratio)?

1 part = 2.4 L (1 mark); water = 7 × 2.4 = 16.8 L (1 mark)

OR

(iii) Lemon juice costs ₹ 60 per litre and sugar ₹ 40 per litre. Find the cost of the ingredients for one 5-litre batch (water is free).

Juice = 0.5 L → ₹ 30; sugar = 1 L → ₹ 40 (1 mark); total ₹ 70 (1 mark)